

Modeling Spatio-Temporal Characteristics of Urban Heat in Bangkok

Juan A. Acero

Vivek K. Sinh

Steven L. Rubinyi



WORLD BANK GROUP

Urban, Disaster Risk Management, Resilience and Land Global Department
June 2025

Abstract

Urban areas accumulate heat, developing distinct urban climates that differ from the regional climate, leading to elevated mean air temperatures within cities. In tropical climates, such as Bangkok, this urban heat can contribute to high levels of heat stress. This study analyzes the spatial and temporal variation of air temperature in the Bangkok Metropolitan Administration (BMA) using dynamic climate modeling (WRF, v4.2). The analysis focuses on three distinct climatic periods: the cool and dry season (November–February), the hot and dry season (March–May), and the wet monsoon season (June–October). Results indicate that during sunrise in the cool and dry season, urban temperatures can be up to 6.4°C higher than those

in surrounding rural areas. The highest temperature differences (>4°C) occur at night during this season, with over 50% of BMA's urban area and population experiencing sustained exposure to these elevated temperatures. In contrast, the smallest temperature differences occur in the hot and dry season, despite it being the hottest overall, due to low soil moisture limiting rural cooling. Under specific conditions, an urban cool island (T_{urban} < T_{rural}) may develop during the daytime. Compact urban areas exhibit the most significant heating, although vegetated areas within BMA are also affected. These findings support the design of targeted mitigation strategies.

This paper is a product of the Urban, Disaster Risk Management, Resilience and Land Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at <http://www.worldbank.org/prwp>. The authors may be contacted at srubinyi@worldbank.org.

The Policy Research Working Paper Series disseminates the findings of work in progress to encourage the exchange of ideas about development issues. An objective of the series is to get the findings out quickly, even if the presentations are less than fully polished. The papers carry the names of the authors and should be cited accordingly. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent.

Modeling Spatio-Temporal Characteristics of Urban Heat in Bangkok*

Juan A. Acero, Vivek K. Sinh, Steven L. Rubinyi

Keywords: urban heat island, climate modeling, heat stress, Bangkok, land use

JEL Classification Codes: **Q54** (Natural Disasters and Their Management); **R14** (Land Use Patterns); **R11** (Regional Economic Activity); **O18** (Urban Policy); **Q51** (Valuation of Environmental Effects)

* This paper was supported by the Global Facility for Disaster Reduction and Recovery (GFDRR). The findings, interpretations, and conclusions expressed in this paper are those of the authors and do not necessarily reflect the views of the World Bank, its affiliated organizations, or GFDRR.

1. Introduction

Hot and humid conditions in tropical metropolitan areas like Bangkok frequently result in high levels of heat stress. Populations in such cities already experience elevated temperatures, which are further exacerbated by humidity, making them highly vulnerable to extreme heat events linked to climate change. As Bangkok's population has increased over recent decades, urban sprawl has expanded, contributing to changes in the local climate (Varnakovida & Ko, 2023). The replacement of natural or rural surfaces with artificial impervious surfaces has a direct effect on urban climate, leading to the development of the Urban Heat Island (UHI) phenomenon—defined as a metropolitan area that is significantly warmer than its surrounding rural areas (Oke, 1987; Oke T. R. et al., 2017; Tian et al., 2021). UHI is characterized not only by increased air temperature—particularly at night—but also by reduced humidity and lower mean wind speeds compared to rural surroundings. Due to the complexity of urban morphology and its interaction with the regional climate, a wide range of microclimates can form within a single metropolitan area (Oke T. R. et al., 2017; Stewart & Oke, 2012).

In addition to its structural characteristics, urbanization introduces anthropogenic heat emissions from sources such as air conditioning, transportation, and industrial activity, which further intensify the UHI effect ((Allegrini et al., 2015; Oke T. R. et al., 2017; Singh et al., 2022). These anthropogenic heat (AH) fluxes from transport, buildings, and industry significantly affect the urban energy balance and contribute to localized warming (Chow et al., 2014; Hii et al., 2014). However, estimating these fluxes remains challenging, and empirical models are commonly used to approximate emissions across sectors (Singh et al., 2020).

Given rising global temperatures (Limsakul, 2020) and already high levels of heat stress in Bangkok (Arifwidodo & Chandrasiri, 2020), quantifying the contribution of urban warming to regional climate dynamics is critical for effective climate adaptation (Webster & McElwee, 2009; World Bank, 2009). Studies from East Asian cities, including Bangkok, have found that mortality increases by 2–6% for every 1°C rise in air temperature above 29°C—a threshold frequently exceeded during daytime in Bangkok. Moreover, the combined effects of heat stress and air pollution—both of which are major concerns in the city—can compound health risks (Areal et al., 2022). The cumulative health impact from simultaneous exposure to both stressors has been shown to exceed their individual effects. For instance, a study in California by Rahman et al. (2022) reported that while extreme temperatures and elevated PM_{2.5} levels individually raised mortality risk by 6.1% and 5.0%, respectively, the combined exposure led to a 21% increase.

As urban climate dynamics drive both UHI and air pollution, integrated mitigation strategies are required. This paper presents a spatio-temporal analysis of the UHI phenomenon in the Bangkok Metropolitan Administration (BMA), focusing on temperature variation across different land-use types and over time under varying regional weather conditions. The findings provide a foundation for evidence-based mitigation strategies aimed at reducing health risks and enhancing urban climate resilience.

2. Methodology

2.1 Study area and its climate

Bangkok Metropolitan Administration (BMA) is located at approximately 13.75°N latitude and 100.50°E longitude, on the delta of the Chao Phraya River, about 40 km from the Gulf of Thailand. The city sits at a mean elevation of 2 meters above sea level and is divided into 50 districts spanning 1,570 km². Over recent decades, Bangkok's urban area has expanded at an average annual rate of ~5% (Kamarajuggedda & Lo, 2019) though this growth has slowed in recent years. BMA is home to over 11 million people, while the wider Bangkok Metropolitan Region (BMR) has a population of around 17 million.

BMA has a tropical savanna climate (Köppen classification: Aw), characterized by hot, wet, and dry conditions. The annual mean temperature is approximately 30 °C, and average annual precipitation is ~1,652 mm (Arifwidodo & Tanaka, 2015). Rainfall occurs on roughly 42% of days throughout the year. The city experiences three main seasons (Pakarnseree et al., 2018; Sintunawa, 2009):

- a) *Cool and dry season (November to February)*: Daily temperatures range from 24 °C to 32 °C. Humidity is at its lowest (~56%), and rainfall is minimal (10–14 mm/month). Cloud cover is also relatively low, with mostly clear skies on 45% of days.
- b) *Hot and dry season (March to May)*: Daily mean temperatures often exceed 32 °C. The highest daily maximum typically occurs in April (35.5 °C), and night-time temperatures remain high, averaging 28.9 °C in May. Rainfall is limited (25–82 mm/month), and relative humidity is moderate (~62%).
- c) *Wet season (June to October)*: Influenced by the Asian monsoon, this period brings heavy rainfall and frequent thunderstorms. The highest monthly rainfall occurs in October (201.5 mm), skies are generally cloudy, and average relative humidity is high (~71%). Daily maximum temperatures reach around 32 °C.

Wind speed and direction vary significantly by season. The wet season is typically the windiest. From June to September, winds have a strong westerly component due to the active southwest monsoon. During the transition period from the southwest to northeast monsoon (September to November), wind speeds are lowest (2.6 m/s in October). Northeasterly winds dominate from December to February. During the hot and dry season (March to May), winds predominantly come from the south and are often influenced by sea breezes (Ngamsiriudom & Tanaka, 2023; Paton & Manomaiphiboon, 2013).

Over the past 50 years, the mean air temperature in BMA has increased by approximately 0.32 °C per decade. Between 1980 and 2012, the annual mean maximum and minimum temperatures rose

by 0.95 °C and 1.97 °C, respectively. Notably, the cool season has become colder, while the hot season has become warmer. Climate projections suggest that by the end of the century, average temperatures in BMA could rise by approximately 2.3 °C under a moderate carbon emission scenario, and up to 3.8 °C under a high emissions scenario (World Bank Group, 2021)

2.2 *UHI Modeling approach*

The effects of urban heat in the Bangkok Metropolitan Administration (BMA) have been assessed using the Weather Research and Forecasting (WRF) model, version 4.2, in combination with the Building Effect Parametrization (BEP) multi-layer canopy model (Martilli et al., 2002) and the Building Energy Model (BEM) (Salamanca et al., 2010).

High-resolution land use and land cover data were employed to provide a realistic representation of urban and natural surface characteristics. The BEM, integrated within the WRF framework, incorporates a range of factors influencing urban energy balance. These include natural ventilation, heat transfer through building materials, indoor radiation exchange, internal heat generation from occupants and equipment, and energy use from air conditioning systems.

Surface energy budgets play a critical role in shaping the outcomes of numerical climate models. To capture the interactions between land surfaces and the urban atmosphere, the NOAH Land Surface Model (LSM) is coupled with WRF to simulate dynamic and thermal processes in the urban canopy (Wang et al., 2018). The NOAH LSM estimates heat fluxes from natural areas, while the BEP/BEM components of the WRF model account for urban heat fluxes. This advanced modeling configuration improves the estimation of surface heat fluxes—both sensible and latent—and enables a more accurate simulation of near-surface air temperatures in urban environments.

2.3 *Climate simulation setup*

To assess the influence of urbanization on urban heat dynamics, the three principal climate seasons in the BMA—the cool and dry season, the hot and dry season, and the wet season (as outlined in Section 2.1)—were each simulated for a representative month: December 2019, March 2020, and July 2020, respectively. These months were selected based on a year that exhibited neutral El Niño–Southern Oscillation (ENSO) conditions to avoid bias from extreme regional climate variability.

The WRF model employed a three-domain nested grid configuration (D1 to D3; see Figure 1), with horizontal grid sizes of 112×112 for both D1 and D2, and 235×350 for D3. The corresponding spatial resolutions were 3 km, 1 km, and approximately 0.3 km, respectively.

Initial and boundary conditions were derived from the Global Data Assimilation System (GDAS) 6-hourly data (GDAS/FNL) at a spatial resolution of 0.25°×0.25°. The model time step was set to 81 seconds for the outermost domain and was reduced by a factor of three for each nested domain.

Vertical resolution included 51 levels extending from the surface up to 50 hPa, with the lowest model level at 5.5 meters above ground. Approximately 28 vertical levels were positioned below 1 km to accurately capture boundary layer processes.

Physical parameterizations included the Kessler microphysics scheme (Kessler, 1969), the RRTM longwave radiation scheme (Mlawer et al., 1997), the Dudhia shortwave radiation scheme (Dudhia, 1989), the Eta similarity surface layer scheme (Janjic Z.L., 1996), the Kain–Fritsch cumulus parameterization (Kain, 2004) and the NOAH land surface scheme (Tewari et al., 2007). The Bougeault–Lacarrere scheme was used to model the planetary boundary layer (Bougeault & Lacarrere, 1989). A 16-hour spin-up period was applied at the start of each simulation to minimize the influence of initial and boundary condition artifacts.

Land use data were drawn from the Local Climate Zone (LCZ) classification for Bangkok, obtained via the World Urban Database and Access Portal Tools (WUDAPT) (Bechtel et al., 2019) (see Figure 2). For spatio-temporal statistical analyses of model outputs, all times were converted from Coordinated Universal Time (UTC) to Local Time (LT) by adding +07:00. All results presented in this paper are referenced in LT.

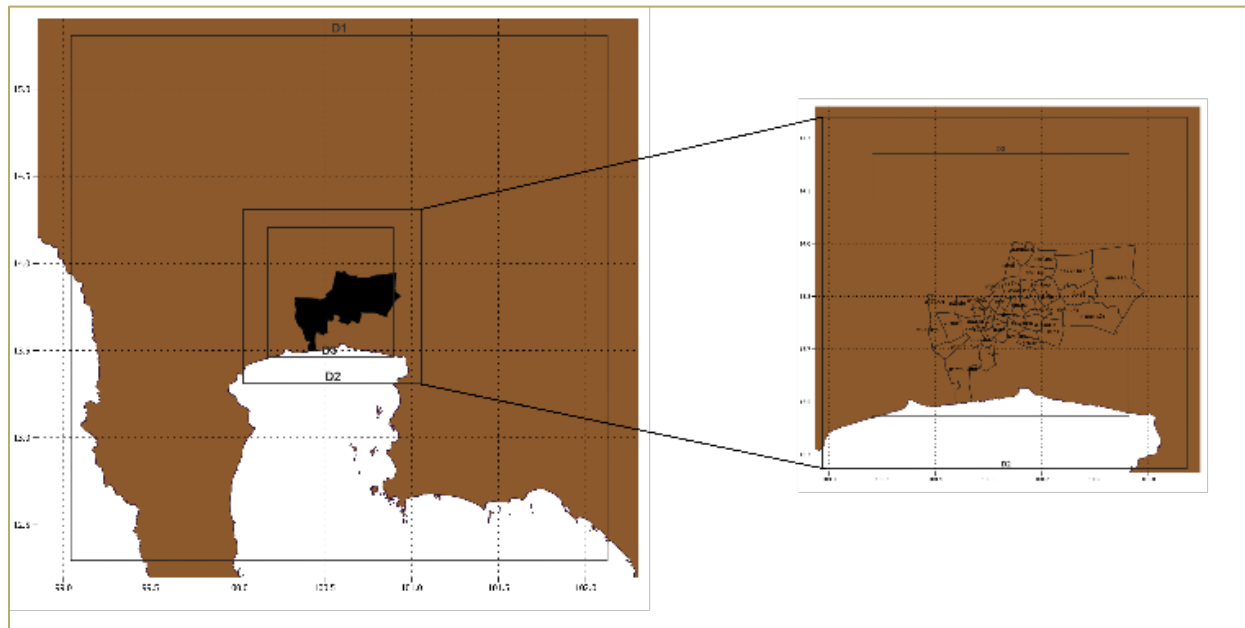


Figure 1. WRF grid configuration for each simulation

temperature and the average air temperature at these reference locations, using the following equation:

$$UHII = T_{a_{simul}} - T_{a_{ref}} \quad (1)$$

where $T_{a_{simul}}$ are the outputs of the model at each grid point and $T_{a_{ref}}$ are the air temperatures at the rural/reference locations.

To assess spatial and seasonal variations in UHII, simulations were conducted for three representative months—December 2019, March 2020, and July 2020—each corresponding to one of Bangkok’s major climatic seasons (as defined in Section 2.1). UHII levels were analyzed across different land use categories, and population exposure estimates were derived accordingly. Given that the UHI phenomenon modifies regional climate conditions, its impact on mean daily maximum and minimum air temperatures in BMA was also evaluated and compared with background rural values.

3. Results

3.1 *Seasonal UHII levels*

BMA experiences three distinct regional climatic patterns: the cool and dry season, the hot and dry season, and the wet monsoon season (as described in Section 2.1). Each season, shaped by its unique meteorological characteristics and interaction with land use patterns, produces differing levels and spatial distributions of UHII throughout the year.

3.1.1 *Cool and dry season*

Mean UHII values for different periods of the day in December 2019 are presented in Figure 3. As shown, levels of urban heat vary significantly throughout the day, consistent with the accumulation of heat in impervious urban materials (Oke, 1982; Oke T. R. et al., 2017). As expected, maximum UHII occurs during the night. Lower UHII levels are observed in the morning, when rural areas heat up more quickly than urban areas due to greater solar access (and thus radiative heating) in open spaces compared to the urban fabric.

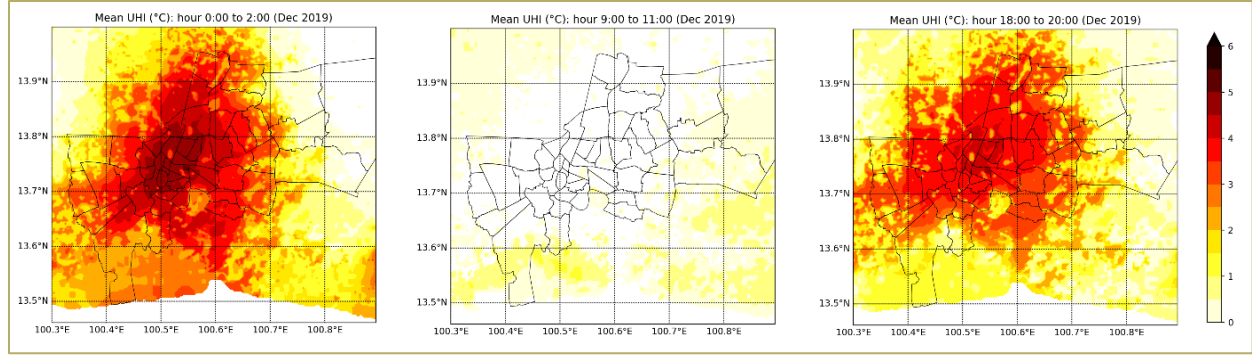


Figure 3. Spatial distribution of UHII levels in the cold and dry season (December 2019) during the a) night 00:00-02:00), morning (09:00-11:00) and evening (18:00-20:00)

During the night (00:00–02:00), the highest UHII levels occur in the central part of BMA. This period shows the highest percentage of area (20.4%) with $\text{UHII} \geq 4^\circ\text{C}$ (Table 1). Out of 50 districts, 27 have a spatial mean $\text{UHII} \geq 4^\circ\text{C}$. The following districts have 100% of their area exceeding this threshold: Pom Prap Sattru Phai, Samphanthawong, Ratchathewi, Phaya Thai, Din Daeng, Bangkok Yai, and Sathon, accounting for 7.9% of the population. The local maximum UHII reaches 5.3°C in Pathum Wan, although only 83.6% of its area has $\text{UHII} \geq 4^\circ\text{C}$. Unlike other districts, the presence of large green areas in Pathum Wan reduced UHII exposure. This inverse relationship between green areas and UHII has been reported in many studies (Gunawardena et al., 2017; Tian et al., 2021). Districts with the lowest spatial mean UHII ($<1^\circ\text{C}$) include Min Buri, Lat Krabang, Khlong Sam Wa, and Nong Chok. In these districts, more than 80% of the area had $\text{UHII} < 2^\circ\text{C}$, and levels never reached 4°C . Despite their low UHII levels, these four districts account for a significant portion of BMA’s population (~12.9%). Nong Chok, located far from the urban core, had the lowest spatial mean UHII ($\sim 0^\circ\text{C}$), with 100% of its area below 2°C . A full breakdown of UHII by district is available in Table A1.1 (Annex 1).

In the morning, the local maximum UHII is the lowest of the day (1.0°C), consistent with patterns observed in other cities (Anjos et al., 2020; Oke, 1982; Soltani & Sharifi, 2017). At this time, most urban areas show UHII levels close to 0°C —similar to other global cities that sometimes experience a “cold urban island” effect (Acero et al., 2013; Bouketta, 2023; Qiu et al., 2023). However, during the evening, as solar intensity decreases, accumulated heat in urban materials (more than in rural areas) causes a rise in BMA’s spatial mean UHII to 2.2°C , with a local maximum of 4.9°C in the central region. In comparison to nighttime values, evening $\text{UHII} \geq 4^\circ\text{C}$ occurs in only 3.2% of the area, and just 5 districts have a spatial mean $\text{UHII} \geq 4^\circ\text{C}$ (Table A1.2, Annex 1). The districts with the lowest UHII remain the same: Min Buri, Lat Krabang, Khlong Sam Wa, and Nong Chok.

Table 1. UHII levels in BMA throughout December 2019 (cool and dry season)

Period	Max UHII	Mean_UHII (Spatial mean)	Area(%) UHI<2°C	Area(%) 2°C≤UHI<4°C	Area(%) UHI≥4°C
Night (Hour: 0:00-2:00)	5.3°C	2.4°C	37.1	42.4	20.4
Morning (Hour: 9:00-11:00)	1.0°C	-0.2°C	100.0	0.0	0.0
Evening (Hour: 18:00-20:00)	4.9°C	2.2°C	45.5	51.3	3.2

3.1.2 Hot and dry season

March is representative of the hot and dry season in BMA. In March 2020, the ENSO phenomenon was in a neutral phase. During this month, UHII levels were lower than in both the cool and dry season (Section 3.1.1) and the wet monsoon period (Section 3.1.3). Additionally, the UHII diurnal cycle shows distinct characteristics. In particular, the highest UHII levels occur in the evening hours, associated with southerly winds transporting urban warm air into the northern region of BMA (Figure 4).

During the hot and dry season, the temperature difference between rural and urban areas tends to diminish. This is due to the reduced cooling capacity of vegetation under very dry conditions in rural areas, which leads to a general decline in UHII levels (Acero & González-Asensio, 2018; Cui & De Foy, 2012; Oke, 1987). Unlike the cool and dry season, the nighttime spatial mean UHII in BMA is only 1.0°C, and most of the area (94.4%) shows mean UHII < 2.0°C. Only three districts—Lak Si, Bang Sue, and Chatuchak, located in the central-northern part of BMA—reach a spatial mean UHII of 2.0°C (Table A1.3 in Annex 1). The local maximum UHII is 2.5°C (Table 2).

During the morning, UHII levels and their spatial distribution resemble those in other seasons. High temperatures and dry conditions in rural areas enhance the intensity of the “cool island” effect—i.e., urban temperatures may be slightly lower than rural temperatures due to reduced radiative heating within the urban fabric. In contrast, UHII levels increase during the evening, with a spatial mean of 1.9°C in BMA and a local maximum of 4.3°C in Don Mueang. Don Mueang also registers the highest spatial mean UHII (3.7°C) during this period (Table A1.4 in Annex 1). Up to eight districts record spatial mean UHII ≥ 3.0°C, comprising 21.6% of BMA’s population. However, UHII levels above 4.0°C remain rare, covering only 1.0% of BMA’s area, primarily in Sai Mai and Don Mueang.

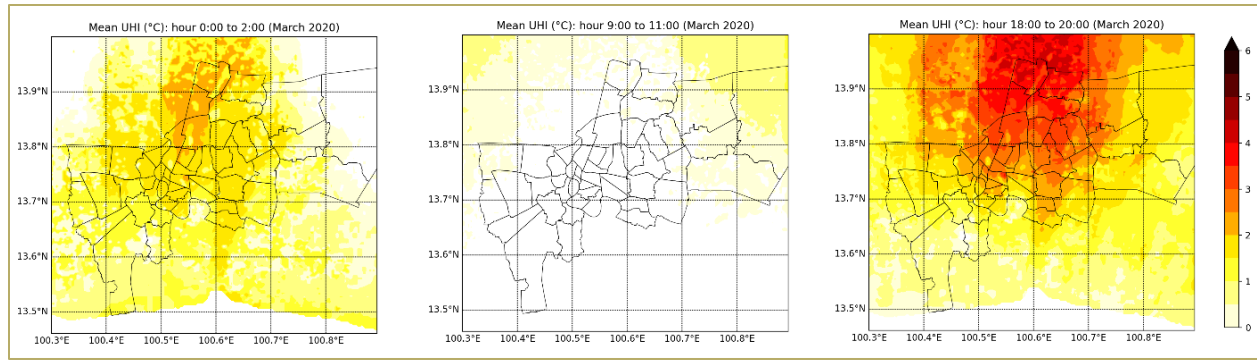


Figure 4. Spatial distribution of UHII levels in the hot and dry season (March 2020) during the a) night 00:00-02:00), morning (09:00-11:00) and evening (18:00-20:00)

Table 2. Mean UHII levels in BMA throughout March 2020 (hot and dry season)

Period	Max UHII	Mean_UHII (Spatial mean)	Area(%) UHI<2°C	Area(%) 2°C≤UHI<4°C	Area(%) UHI≥4°C
Night (Hour: 0:00-2:00)	2.5°C	1.0°C	94.4	5.6	0.0
Morning (Hour: 9:00-11:00)	1.1°C	-0.4°C	100.0	0.0	0.0
Evening (Hour: 18:00-20:00)	4.3°C	1.9°C	59.6	39.5	1.0

Although the interaction between the regional climate and metropolitan area during the hot and dry season results in the lowest UHII levels, the mean air temperatures during this period are the highest of the year. This is driven by low precipitation and frequent clear skies characteristic of the inter-monsoon period (see Section 3.5). In BMA, low UHII levels do not correspond to lower heat stress; in fact, heat stress levels are typically highest during the hot and dry season.

3.1.3. Wet monsoon season

July 2020 is selected as a representative month of the wet monsoon season in BMA, characterized by high levels of rainfall. UHII levels and their diurnal pattern in this period are similar to those observed during the cool and dry season (Section 3.1.1).

During the night, UHII levels are highest in the central part of BMA, with a local maximum of 3.9°C in Pathum Wan district (Figure 5). Half of BMA is exposed to mean UHII levels below 2°C, while the other half falls between 2°C and 4°C (Table 3). The spatial mean UHII in BMA is 1.7°C. However, 15 districts in the central area of BMA register spatial mean UHII levels slightly above 3°C, comprising 18.5% of the population. These districts are listed in Table A1.5 in Annex 1. As in the cool and dry season, the districts with the lowest spatial mean UHII (<1°C) are Min Buri, Lat Krabang, Khlong Sam Wa, and Nong Chok.

During the morning, the local maximum UHII in BMA is 1.6°C—higher than in other seasons, though still much lower than the night and evening values. The spatial mean does not indicate a “cool island” effect during the morning in July, primarily due to high soil moisture resulting from frequent and abundant rainfall. Under these conditions, rural areas maintain lower air temperatures than urban zones (Oke, 1987). In the evening, UHII levels rise again, with a BMA spatial mean of 1.5°C, approaching nighttime levels. However, only 38.9% of the area records UHII values above 2°C, and no districts register spatial mean UHII $\geq 3^\circ\text{C}$. As in the night period of the cool and dry season, the lowest UHII levels occur in Min Buri, Lat Krabang, Khlong Sam Wa, and Nong Chok.

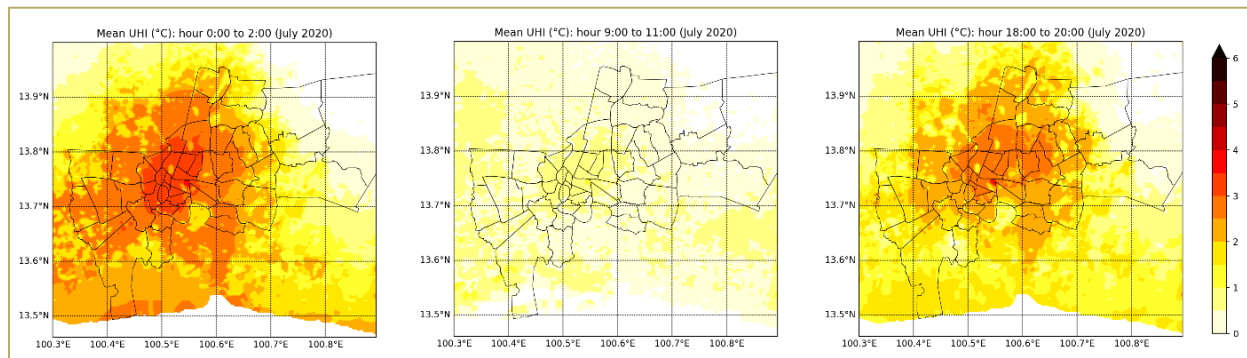


Figure 5. Spatial distribution of UHII levels in the wet monsoon season (July 2020) during the a) night 00:00-02:00), morning (09:00-11:00) and evening (18:00-20:00)

Table 3. Mean UHII levels in BMA throughout July 2020 (wet monsoon season)

Period	Max UHII	Mean_UHII (Spatial mean)	Area(%) UHI<2°C	Area(%) 2°C≤UHI<4°C	Area(%) UHI≥4°C
Night (Hour: 0:00-2:00)	3.9°C	1.7°C	49.4	50.6	0.0
Morning (Hour: 9:00-11:00)	1.6°C	0.3°C	100.0	0.0	0.0
Evening (Hour: 18:00-20:00)	3.9°C	1.5°C	61.1	38.9	0.0

3.2 Land use and UHII levels

As presented in Section 3.1, the spatial distribution of UHII varies across seasons based on the interaction between land use and regional climate. Figure 6 shows the mean UHII levels across different land use categories, influenced by urban morphology, surface materials, and vegetation cover—all of which affect the capacity to absorb heat during the day and release it at night.

Mean UHII levels are highest in the cool and dry season (December) and lowest in the hot and dry season (March), primarily driven by prevailing regional weather conditions. As illustrated in Figure 6, this seasonal pattern is consistent across all land use categories in BMA. The highest

levels of overheating are observed in compact high-rise developments, where daily mean UHII reaches 3.6°C in December. Other compact urban areas with limited greenery also exhibit elevated UHII levels. In BMA, as building massing decreases (i.e., areas become less compact and buildings are lower), UHII levels tend to decrease.

This finding differs slightly from some studies, which report that compact mid-rise developments experience the highest UHII levels (Rahmani & Sharifi, 2025). These differences can be attributed to local urban characteristics, latitude, and climatic conditions (Oke, 1987; Oke T. R. et al., 2017). Similar to findings in other cities, areas with more vegetation—such as open low-rise developments—exhibit lower UHII levels. A recent study in Beijing (Zheng et al., 2024) similarly found that vegetated areas recorded the lowest UHII intensities. Naturally vegetated areas within BMA, such as croplands and forests, also experience some urban heat influence. Atmospheric circulations can advect warm air from overheated urban cores into these natural zones, with mean UHII levels in such areas reaching up to 1°C.

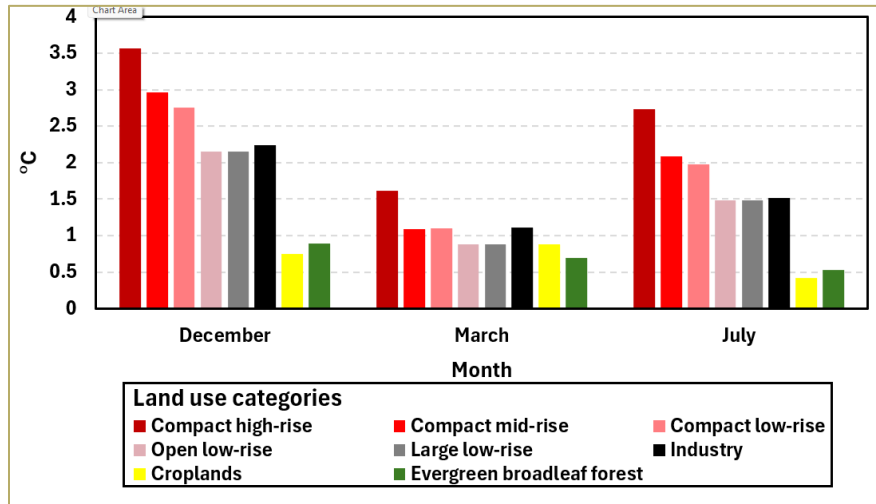


Figure 6. Mean UHI intensity levels for each type of land use in the three climatic seasons in BMA

3.3 Mean diurnal UHII cycle based on land use category

In addition to seasonal variation in mean UHII levels, the diurnal pattern of UHII also varies across the year (Section 3.1). The three climatic seasons in BMA exhibit distinct hourly UHII dynamics, with the cool and dry season (December) displaying the largest diurnal amplitude (Figure 7).

As discussed in Section 3.1, maximum mean UHII levels during December are recorded just before sunrise in compact high-rise areas, reaching 6.3°C in 2019 (Figure 7). After sunrise, UHII levels drop significantly. During the morning and throughout the afternoon, the temperature difference between urban and rural areas becomes negligible. On average, a modest urban cool island effect ($T_{\text{urban}} < T_{\text{rural}}$) can emerge due to reduced solar radiation penetration into the urban

canopy and a deeper urban boundary layer (Theeuwes et al., 2015). In December 2019, temperatures in low- and mid-rise developments were up to 0.6°C cooler than rural reference sites during the day, consistent with findings in other cities. Industrial areas showed a more pronounced cool island (−0.8°C), attributed to their central location within BMA and prevailing atmospheric conditions during early morning hours (Theeuwes et al., 2015).

In the hot and dry season (March), maximum mean UHII occurs in the evening (19:00 LT), reaching 3.3°C in compact high-rise zones (Figure 7). Pre-sunrise UHII levels are slightly lower, at 2.8°C. During the wet season (July), maximum values after sunset and before sunrise are 3.7°C and 3.8°C, respectively. The dry background climate and very low soil moisture in March not only result in the lowest daily mean UHII levels of the year, but also intensify the urban cool island effect. In certain areas, this effect reaches −1.5°C, as rural surfaces heat more rapidly than urban ones, where dense built structures limit solar access (Acero & González-Asensio, 2018; Rasul et al., 2015). Similar negative UHII levels—most pronounced in the mid-afternoon (around 16:00 LT)—have been reported in previous studies on BMA (Kamma et al., 2017). with cloud cover also identified as a contributing factor. Consistent with our results and Kamma et al. (2017), Theeuwes et al. (2015) observed that urban cool islands in mid-latitudes typically peak four hours after sunrise and can persist into the early afternoon.

Vegetated areas within BMA are also affected by urban heat dynamics. While their UHII values before sunrise are lower than those of impervious urban zones (peaking at 1.6°C), they tend to show negligible UHII during daytime hours.

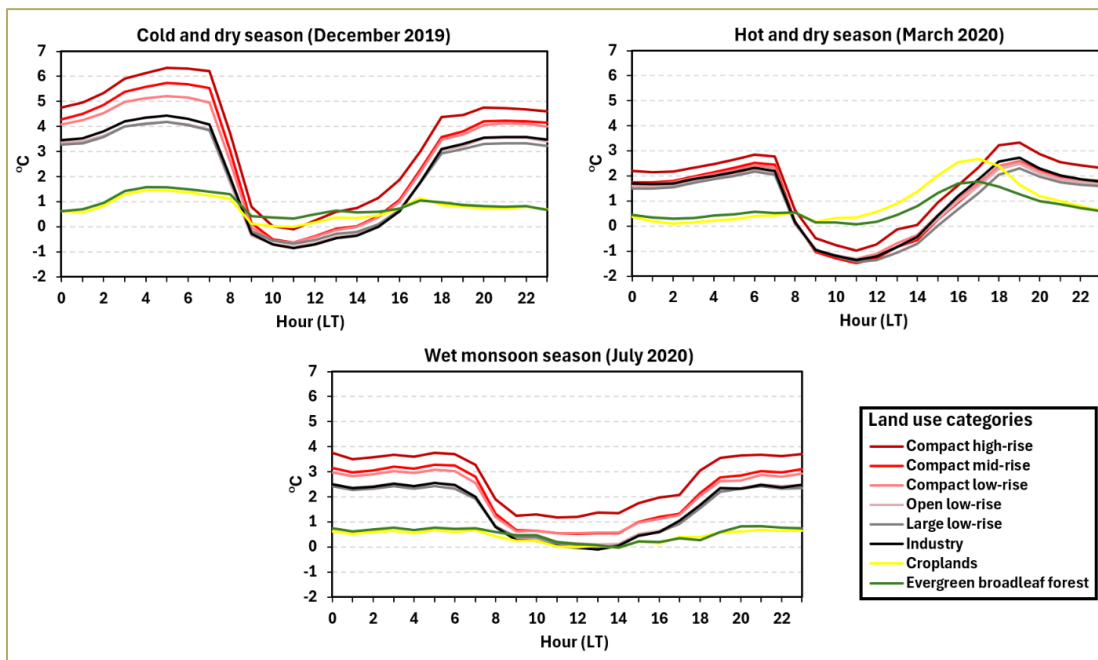


Figure 7. Mean diurnal cycle of UHI intensity for each type of land use in the three climatic seasons in BMA

3.4 *Exposure of population to urban heat*

Annex 1 presents the percentage of BMA's population residing in each district. A comparison between UHII levels and population shares across districts reveals no clear correlation, underscoring the importance of district location and land use characteristics in shaping UHII outcomes.

As with the diurnal variability of UHII (Section 3.3), the proportion of the population exposed to different UHII levels also fluctuates throughout the day. The most severe urban heat exposure occurs during the evening and night-time hours (Figure 7). From morning to late afternoon, the majority of the population is exposed to UHII levels below 2°C (Figure 8). However, in the evening, up to 90.8% of the urbanized area (in December 2019) experiences UHII levels between 2–4°C. At night-time, more than half (54.6%) of the urban area sees UHII levels exceeding 4°C—posing significant implications for human health (Baccini et al., 2008; Gasparrini et al., 2015; Heaviside et al., 2017) and energy demand (Hirano & Fujita, 2012; Li et al., 2019).

High night-time temperatures can impair the body's ability to recover from heat stress (Kim et al., 2023; Murage et al., 2017). Maintaining comfortable indoor thermal conditions often requires air conditioning, resulting in increased energy consumption. Therefore, elevated night-time UHII not only exacerbates heat-related health risks but also amplifies energy use, with broader economic repercussions. This complements evidence that high air temperatures negatively affect daytime outdoor thermal comfort in BMA (Arifwidodo & Chandrasiri, 2020).

Consistent with seasonal UHII patterns (Section 3.1), the hot and dry (March) and wet monsoon (July) seasons expose the population to lower UHII levels despite having higher ambient air temperatures than the cool and dry season (December) (see Section 3.5). In March, only 5.8% of the BMA population is exposed to UHII levels above 4°C during the evening. In July, evening and night-time UHII values are moderate, but approximately 85% of the population still experiences UHII between 2–4°C.

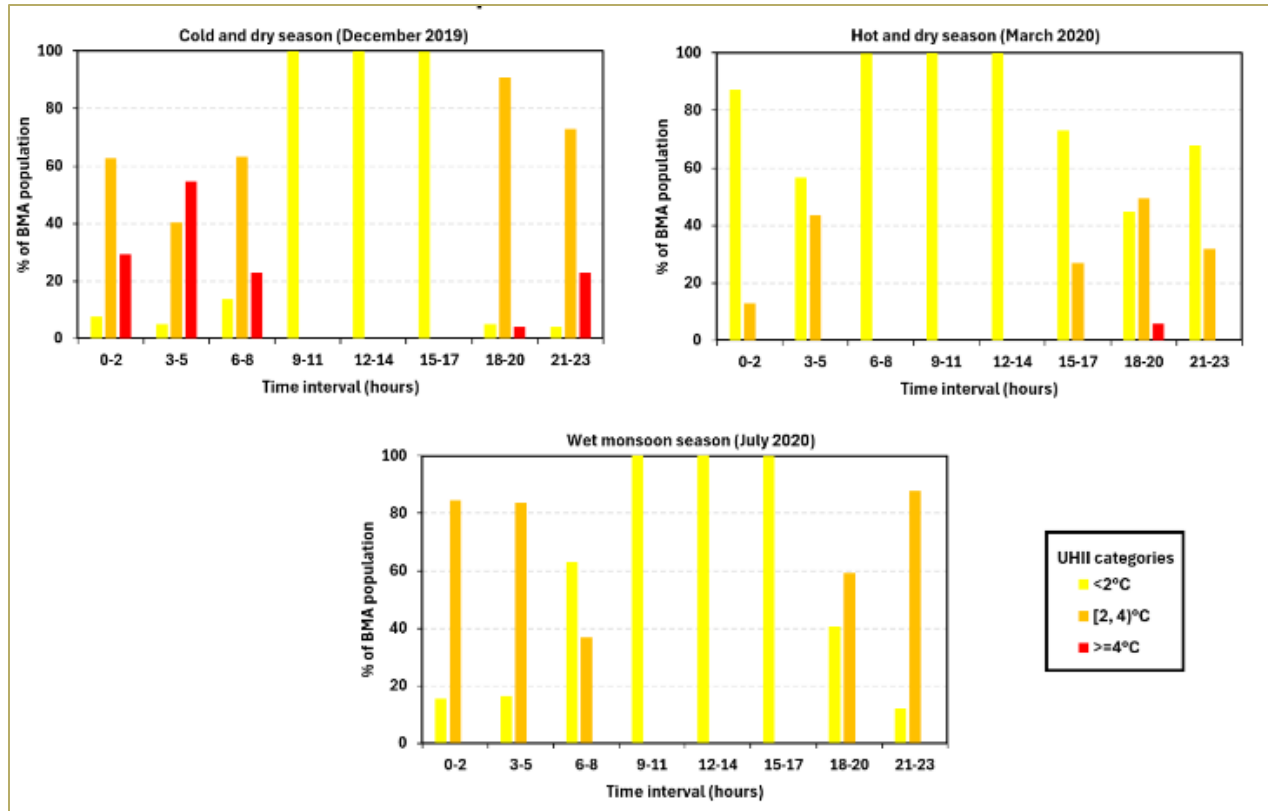


Figure 8. Estimation of BMA population (based on urbanized area in each district) exposed to different ranges of UHII

3.5 Variation of regional climate

UHII levels in BMA vary across time of day, season, and district. Importantly, urban heat modifies the background regional climate. Table 4 presents daily maximum temperature ($T_{\max}$), daily minimum temperature ($T_{\min}$) and mean temperature (T_{mean}) for a rural reference environment and the urban area of BMA across different climatic seasons.

Consistent with the seasonal and diurnal patterns described in Sections 3.1 and 3.3, the impact of urban heat is most pronounced for $T_{\min}$ (typically observed around sunrise: 6:00–8:00), with mean values increasing by 6°C relative to rural sites during the wet monsoon season (July). The lowest impact on $T_{\min}$ occurs during the hot and dry season (March), aligning with low soil moisture and reduced cooling potential from dry vegetation in rural areas at night (Oke, 1987, 1989).

In contrast, $T_{\max}$ (typically recorded in the afternoon) shows a more modest urban influence, with a mean increase of 0.5°C during both the wet monsoon and the cool and dry seasons. During the hot and dry season, $T_{\max}$ in compact urban developments can even be slightly lower than in rural sites (see Section 3.3).

Figure 9 illustrates the spatial distribution of air temperature across BMA, shaped by the interaction of urban form and prevailing weather conditions. Despite UHII levels being lowest

in March (the hot and dry season), $T_{\max}$ reaches highest values ($>34^{\circ}\text{C}$). Note that the values correspond to the time when the rural sites registered the daily minimum ($T_{\min}$) and maximum ($T_{\max}$) mean temperature.

Such thermal conditions can result in significant outdoor heat stress and discomfort (Arifwidodo & Chandrasiri, 2020; Jitkhajornwanich et al., 1998; Srivanit & Jareemit, 2020).

Conversely, even though UHII levels are higher in the cool and dry season (December), absolute air temperatures in urban areas are lower ($T_{\max} = 31.5^{\circ}\text{C}$ and $T_{\min} = 23.9^{\circ}\text{C}$) compared to other seasons (Table 4). This confirms that the highest UHII levels in BMA do not coincide with the highest absolute air temperatures or peak heat stress conditions.

Table 4. Mean UHII levels in BMA throughout March 2020 (hot and dry season)

	Cold & dry season (December 2019)			Hot & dry season (March 2020)			Wet monsoon season (July 2020)		
	$T_{\max}$ ($^{\circ}\text{C}$)	$T_{\min}$ ($^{\circ}\text{C}$)	T_{mean} ($^{\circ}\text{C}$)	$T_{\max}$ ($^{\circ}\text{C}$)	$T_{\min}$ ($^{\circ}\text{C}$)	T_{mean} ($^{\circ}\text{C}$)	$T_{\max}$ ($^{\circ}\text{C}$)	$T_{\min}$ ($^{\circ}\text{C}$)	T_{mean} ($^{\circ}\text{C}$)
Rural/ Reference	31.0	18.8	25.0	34.4	24.8	28.6	32.6	22.3	28.2
Compact urban	31.5	23.9	27.8	34.0	27.1	29.7	33.2	28.3	30.4
Difference	0.5	5.1	2.8	-0.4	2.4	1.1	0.6	6.0	2.0

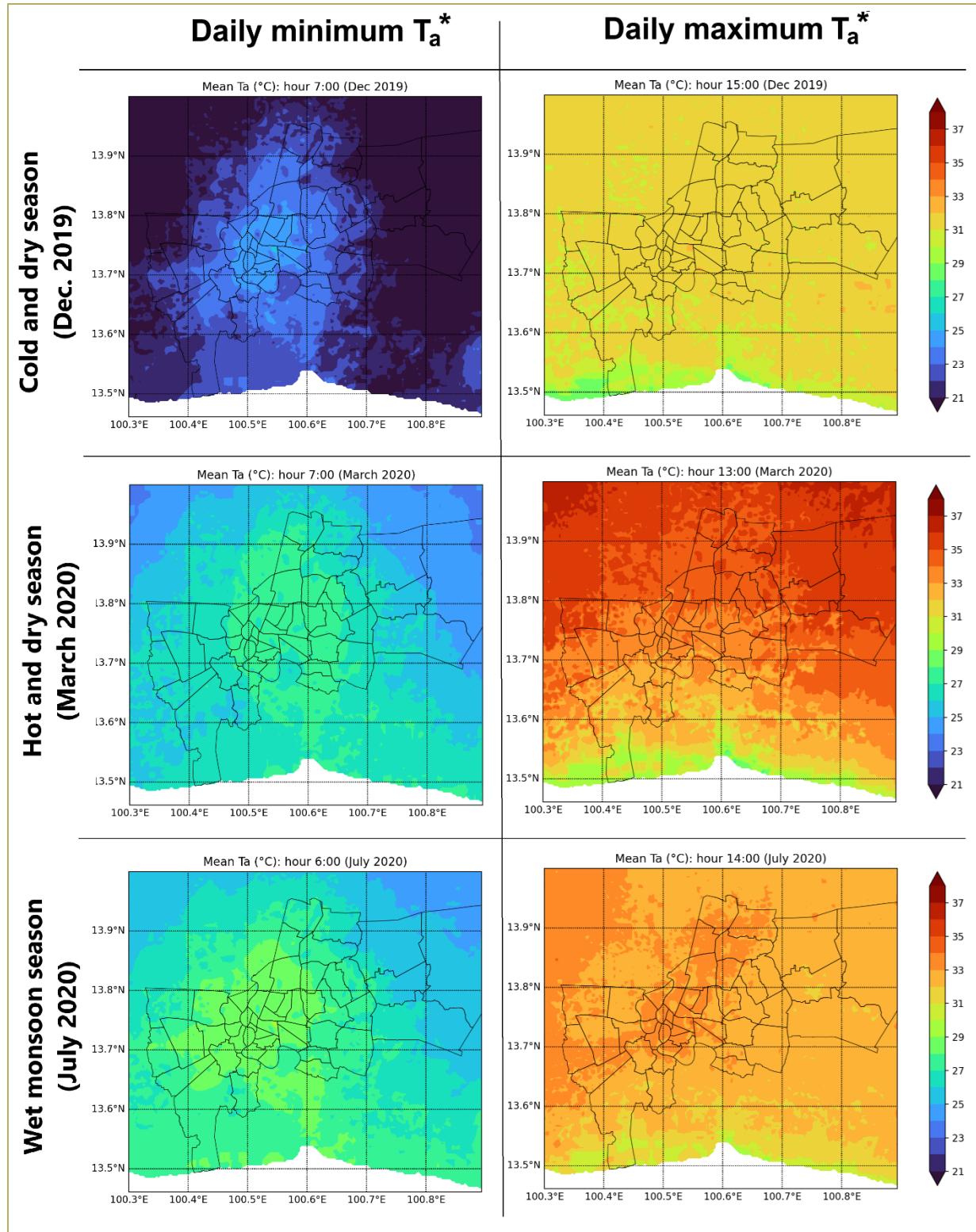


Figure 9. Spatial distribution of mean daily maximum and minimum air temperature for 3 climatic seasons in BMA.

4. Conclusion

This study has examined the spatio-temporal dynamics of the UHI effect in the BMA area using high-resolution dynamic climate modeling. The analysis reveals significant seasonal and diurnal variation in UHII, shaped by the interaction between urban morphology, land use characteristics, and prevailing regional climate conditions. Three major climatic periods were assessed—cool and dry, hot and dry, and wet monsoon seasons—each exhibiting distinct spatial patterns and magnitudes of urban heat intensity.

The most pronounced UHI effects were observed during the cool and dry season (December 2019), when mean UHII levels reached up to 6.4°C at sunrise, and over 50% of the urban population experienced nighttime UHII values exceeding 4°C. These elevated nighttime temperatures are particularly concerning due to their association with increased health risks and reduced opportunities for thermal recovery during sleep. In contrast, during the hot and dry season (March 2020), UHII values were lower due to reduced nighttime cooling in rural areas caused by low soil moisture. Nevertheless, this season was characterized by the highest overall air temperatures and thus the greatest levels of heat stress—highlighting that low UHII does not necessarily equate to low risk.

Daytime UHII levels were generally lower across all seasons, and under certain conditions, an urban cool island effect ($T_{\text{urban}} < T_{\text{rural}}$) was observed, particularly in the hot and dry season. This phenomenon was influenced by solar radiation dynamics and the reduced capacity of impervious surfaces to heat up during early hours when shaded by surrounding structures.

Land use patterns played a significant role in determining UHII intensity. Compact high-rise developments consistently exhibited the highest UHII values, while low-rise and vegetated areas displayed comparatively lower levels. However, even natural areas within BMA registered UHII levels of approximately 0.5–1.0°C, likely due to the advection of heat from surrounding urban zones. These findings underscore that urban heat impacts are not confined to built-up zones but can affect the broader metropolitan landscape.

Beyond documenting the magnitude and distribution of UHI effects, this study contributes to a deeper understanding of the physical mechanisms driving urban heat in tropical megacities. Importantly, it demonstrates that urban heat modifies not only absolute temperature levels but also alters the temporal structure of temperature cycles, particularly minimum temperatures during early morning hours.

These insights provide a strong empirical basis for the development of targeted UHI mitigation strategies. Policy responses should consider the role of urban morphology, prioritize greening in high-exposure areas, and account for the seasonal timing of interventions. As Bangkok continues to urbanize and global temperatures rise, addressing UHI through data-driven urban planning and climate-responsive design will be critical to safeguarding public health, enhancing liveability, and increasing the city's climate resilience.

5. References

- Acero, J. A., Arrizabalaga, J., Kupski, S., & Katschner, L. (2013). Urban heat island in a coastal urban area in northern Spain. *Theoretical and Applied Climatology*, 113(1–2), 137–154. <https://doi.org/10.1007/s00704-012-0774-z>
- Acero, J. A., & González-Asensio, B. (2018). Influence of vegetation on the morning land surface temperature in a tropical humid urban area. *Urban Climate*, 26(August 2017), 231–243. <https://doi.org/10.1016/j.uclim.2018.09.004>
- Addas, A., Goldblatt, R., & Rubinyi, S. (2020). Utilizing Remotely Sensed Observations to Estimate the Urban Heat Island Effect at a Local Scale: Case Study of a University Campus. *Land*, 9(6), 191. <https://doi.org/10.3390/land9060191>
- Allegrini, J., Dorer, V., & Carmeliet, J. (2015). Influence of morphologies on the microclimate in urban neighbourhoods. *Journal of Wind Engineering and Industrial Aerodynamics*, 144, 108–117. <https://doi.org/10.1016/j.jweia.2015.03.024>
- Anjos, M., Targino, A. C., Krecl, P., Oukawa, G. Y., & Braga, R. F. (2020). Analysis of the urban heat island under different synoptic patterns using local climate zones. *Building and Environment*, 185, 107268. <https://doi.org/10.1016/J.BUILDENV.2020.107268>
- Areal, A. T., Zhao, Q., Wigmann, C., Schneider, A., & Schikowski, T. (2022). The effect of air pollution when modified by temperature on respiratory health outcomes: A systematic review and meta-analysis. In *Science of the Total Environment* (Vol. 811). <https://doi.org/10.1016/j.scitotenv.2021.152336>
- Arifwidodo, S. D., & Chandrasiri, O. (2020). Urban heat stress and human health in Bangkok, Thailand. *Environmental Research*, 185. <https://doi.org/10.1016/j.envres.2020.109398>
- Arifwidodo, S. D., & Tanaka, T. (2015). The Characteristics of Urban Heat Island in Bangkok, Thailand. *Procedia - Social and Behavioral Sciences*, 195. <https://doi.org/10.1016/j.sbspro.2015.06.484>
- Baccini, M., Biggeri, A., Accetta, G., Kosatsky, T., Katsouyanni, K., Analitis, A., Anderson, H. R., Bisanti, L., D’Ippoliti, D., Danova, J., Forsberg, B., Medina, S., Paldy, A., Rabchenko, D., Schindler, C., & Michelozzi, P. (2008). Heat effects on mortality in 15 European cities. *Epidemiology*, 19(5). <https://doi.org/10.1097/EDE.0b013e318176bfcd>
- Bechtel, B., Demuzere, M., Mills, G., Zhan, W., Sismanidis, P., Small, C., & Voogt, J. (2019). SUHI analysis using Local Climate Zones—A comparison of 50 cities. *Urban Climate*, 28. <https://doi.org/10.1016/j.uclim.2019.01.005>

- Bougeault, P., & Lacarrere, P. (1989). *Parameterization of Orography-Induced Turbulence in a Meso-beta-Scale Model*. 117, 1872–1890. [https://doi.org/https://doi.org/10.1175/1520-0493\(1989\)117<1872:POOITI>2.0.CO;2](https://doi.org/https://doi.org/10.1175/1520-0493(1989)117<1872:POOITI>2.0.CO;2)
- Bouketta, S. (2023). Urban Cool Island as a sustainable passive cooling strategy of urban spaces under summer conditions in Mediterranean climate. *Sustainable Cities and Society*, 99, 104956. <https://doi.org/10.1016/J.SCS.2023.104956>
- Chow, W. T. L., Salamanca, F., Georgescu, M., Mahalov, A., Milne, J. M., & Ruddell, B. L. (2014). A multi-method and multi-scale approach for estimating city-wide anthropogenic heat fluxes. *Atmospheric Environment*, 99, 64–76. <https://doi.org/10.1016/j.atmosenv.2014.09.053>
- Cui, Y. Y., & De Foy, B. (2012). Seasonal variations of the urban heat island at the surface and the near-surface and reductions due to urban vegetation in Mexico City. *Journal of Applied Meteorology and Climatology*, 51(5). <https://doi.org/10.1175/JAMC-D-11-0104.1>
- Dudhia, J. (1989). *Numerical Study of Convection Observed during the Winter Monsoon Experiment Using a Mesoscale Two-Dimensional Model*. VOL. 46, N, 3077–3107. [https://doi.org/https://doi.org/10.1175/1520-0469\(1989\)046<3077:NSOCOD>2.0.CO;2](https://doi.org/https://doi.org/10.1175/1520-0469(1989)046<3077:NSOCOD>2.0.CO;2)
- Gasparrini, A., Guo, Y., Hashizume, M., Lavigne, E., Zanobetti, A., Schwartz, J., Tobias, A., Tong, S., Rocklöv, J., Forsberg, B., Leone, M., De Sario, M., Bell, M. L., Guo, Y. L. L., Wu, C. F., Kan, H., Yi, S. M., De Sousa Zanotti Stagliorio Coelho, M., Saldiva, P. H. N., ... Armstrong, B. (2015). Mortality risk attributable to high and low ambient temperature: A multicountry observational study. *The Lancet*, 386(9991). [https://doi.org/10.1016/S0140-6736\(14\)62114-0](https://doi.org/10.1016/S0140-6736(14)62114-0)
- Goldblatt, R., Addas, A., Crull, D., Maghrabi, A., Levin, G. G., & Rubinyi, S. (2021). Remotely Sensed Derived Land Surface Temperature (LST) as a Proxy for Air Temperature and Thermal Comfort at a Small Geographical Scale. *Land*, 10(4), 410. <https://doi.org/10.3390/land10040410>
- Gunawardena, K. R. R., Wells, M. J. J., & Kershaw, T. (2017). Utilising green and bluespace to mitigate urban heat island intensity. *Science of the Total Environment*, 584–585, 1040–1055. <https://doi.org/10.1016/j.scitotenv.2017.01.158>
- Heaviside, C., Macintyre, H., & Vardoulakis, S. (2017). The Urban Heat Island: Implications for Health in a Changing Environment. In *Current environmental health reports* (Vol. 4, Issue 3). <https://doi.org/10.1007/s40572-017-0150-3>
- Hii, D., Hii, D., Chung, J., Hien, W. N., & Jusuf, S. K. (2014). *Anthropogenic heat contribution to air temperature increase at pedestrian height in Singapore 's high density Central Business District (CBD). July*, 1–6.

- Hirano, Y., & Fujita, T. (2012). Evaluation of the impact of the urban heat island on residential and commercial energy consumption in Tokyo. *Energy*, 37(1). <https://doi.org/10.1016/j.energy.2011.11.018>
- Janjic Z.L. (1996). The surface layer in the NCEP Eta Model. *Proceedings of the 11th Conference on Numerical Weather Prediction*.
- Jitkhajornwanich, K., Pitts, A. C., Malama, A., & Sharples, S. (1998). Thermal comfort in transitional spaces in the cool season of Bangkok. *ASHRAE Transactions*, 104(Pt 1B).
- Kain, J. S. (2004). The Kain–Fritsch Convective Parameterization: An Update. *JOURNAL OF APPLIED METEOROLOGY*, 43, 12.
- Kamarajugedda, S. A., & Lo, E. Y. M. (2019). MODELLING URBAN GROWTH for BANGKOK and ASSESSING LINKAGES with ROAD DENSITY and SOCIO-ECONOMIC INDICATORS. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives*, 42(4/W19). <https://doi.org/10.5194/isprs-archives-XLII-4-W19-255-2019>
- Kamma, J., Manomaiphiboon, K., & Aman, N. (2017). An Investigation of Urban Cool Island in Bangkok and Its Possible Causes. *The 4th International Environment Asia International Conference*.
- Kessler, E. (1969). On the Distribution and Continuity of Water Substance in Atmospheric Circulations. In *On the Distribution and Continuity of Water Substance in Atmospheric Circulations*. https://doi.org/10.1007/978-1-935704-36-2_1
- Kim, S. E., Hashizume, M., Armstrong, B., Gasparrini, A., Oka, K., Hijioka, Y., Vicedo-Cabrera, A. M., & Honda, Y. (2023). Mortality Risk of Hot Nights: A Nationwide Population-Based Retrospective Study in Japan. *Environmental Health Perspectives*, 131(5). <https://doi.org/10.1289/EHP11444>
- Li, X., Zhou, Y., Yu, S., Jia, G., Li, H., & Li, W. (2019). Urban heat island impacts on building energy consumption: A review of approaches and findings. In *Energy* (Vol. 174). <https://doi.org/10.1016/j.energy.2019.02.183>
- Limsakul, A. (2020). Trends in Thailand's extreme temperature indices during 1955-2018 and their relationship with global mean temperature change. *Applied Environmental Research*, 42(2). <https://doi.org/10.35762/AER.2020.42.2.8>
- Martilli, A., Clappier, A., & Rotach, M. W. (2002). An Urban Surface Exchange Parameterisation for Mesoscale Models. *Boundary-Layer Meteorology*, 104(2), 261–304. <https://doi.org/10.1023/A:1016099921195>
- Mlawer, E. J., Taubman, S. J., Brown, P. D., Iacono, M. J., & Clough, S. A. (1997). Radiative transfer for inhomogeneous atmospheres: RRTM, a validated correlated-k model for the

- longwave. *Journal of Geophysical Research: Atmospheres*, 102(D14), 16663–16682.
<https://doi.org/10.1029/97JD00237>
- Murage, P., Hajat, S., & Sari Kovats, R. (2017). Effect of night-time temperatures on cause and age-specific mortality in London. *Environmental Epidemiology*, 1(2).
<https://doi.org/10.1097/EE9.0000000000000005>
- Ngamsiriudom, T., & Tanaka, T. (2023). Making an urban environmental climate map of the Bangkok Metropolitan Region, Thailand: Analysis of air temperature, wind distributions, and spatial environmental factors. *World Development Sustainability*, 3, 100105.
<https://doi.org/10.1016/J.WDS.2023.100105>
- Oke, T. R. (1982). The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455), 1–24. <https://doi.org/10.1002/qj.49710845502>
- Oke, T. R. (1987). *Boundary Layer Climates*. Routledge.
- Oke, T. R. (1989). The micrometeorology of the urban forest. *Philosophical Transactions - Royal Society of London, B*, 324(1223). <https://doi.org/10.1098/rstb.1989.0051>
- Oke T. R., Mills G., Christen A., & Voogt J. A. (2017). *Urban Climates*. Cambridge University Press.
- Pakarnseree, R., Chunkao, K., & Bualert, S. (2018). Physical characteristics of Bangkok and its urban heat island phenomenon. *Building and Environment*, 143.
<https://doi.org/10.1016/j.buildenv.2018.07.042>
- Paton, C. P., & Manomaiphiboon, K. (2013). A Metropolitan Wind Resource Assessment for Bangkok, Thailand Part 1: Wind Resource Mapping. *Journal of Sustainable Energy & Environment*, 4.
- Qiu, J., Li, X., & Qian, W. (2023). Optimizing the spatial pattern of the cold island to mitigate the urban heat island effect. *Ecological Indicators*, 154, 110550.
<https://doi.org/10.1016/J.ECOLIND.2023.110550>
- Rahman, M. M., McConnell, R., Schlaerth, H., Ko, J., Silva, S., Lurmann, F. W., Palinkas, L., Johnston, J., Hurlburt, M., Yin, H., Ban-Weiss, G., Garcia, E., & Gern, D. (2022). The Effects of Coexposure to Extremes of Heat and Particulate Air Pollution on Mortality in California Implications for Climate Change. *American Journal of Respiratory and Critical Care Medicine*, 206(9). <https://doi.org/10.1164/rccm.202204-0657OC>
- Rahmani, N., & Sharifi, A. (2025). Urban heat dynamics in Local Climate Zones (LCZs): A systematic review. *Building and Environment*, 267, 112225.
<https://doi.org/10.1016/J.BUILDENV.2024.112225>

- Rasul, A., Balzter, H., & Smith, C. (2015). Spatial variation of the daytime Surface Urban Cool Island during the dry season in Erbil, Iraqi Kurdistan, from Landsat 8. *Urban Climate*, 14, 176–186. <https://doi.org/10.1016/j.uclim.2015.09.001>
- Rubinyi, S., Wibisana, P. S., Park, J., Jones, N. K. W., Acero, J. A., & Moeller, P. (2025). *Shaping a cooler Bangkok: Tackling urban heat for a more livable city* (English). The World Bank.
- Salamanca, F., Krpo, A., Martilli, A., & Clappier, A. (2010). A new building energy model coupled with an urban canopy parameterization for urban climate simulations—part I. formulation, verification, and sensitivity analysis of the model. *Theoretical and Applied Climatology*, 99(3–4), 331–344. <https://doi.org/10.1007/s00704-009-0142-9>
- Singh, V. K., Acero, J. A., & Martilli, A. (2020). Evaluation of the impact of anthropogenic heat emissions generated from road transportation and power plants on the UHI intensity of Singapore. *Technical Report Cooling Singapore*, 500.
- Singh, V. K., Mughal, M. O., Martilli, A., Acero, J. A., Ivanchev, J., & Norford, L. K. (2022). Numerical analysis of the impact of anthropogenic emissions on the urban environment of Singapore. *Science of The Total Environment*, 806, 150534. <https://doi.org/10.1016/j.scitotenv.2021.150534>
- Sintunawa, C. (2009). *Bangkok Assessment Report on Climate Change 2009*.
- Soltani, A., & Sharifi, E. (2017). Daily variation of urban heat island effect and its correlations to urban greenery: A case study of Adelaide. *Frontiers of Architectural Research*, 6(4), 529–538. <https://doi.org/10.1016/J.FOAR.2017.08.001>
- Srivanit, M., & Jareemit, D. (2020). Modeling the influences of layouts of residential townhouses and tree-planting patterns on outdoor thermal comfort in Bangkok suburb. *Journal of Building Engineering*, 30. <https://doi.org/10.1016/j.job.2020.101262>
- Stewart, I. D., & Oke, T. R. (2012). Local Climate Zones for Urban Temperature Studies. *Bulletin of the American Meteorological Society*, 93(12), 1879–1900. <https://doi.org/10.1175/BAMS-D-11-00019.1>
- Taha, H. (1997). Urban climates and heat islands: Albedo, evapotranspiration, and anthropogenic heat. *Energy and Buildings*, 25(2), 99–103. [https://doi.org/10.1016/s0378-7788\(96\)00999-1](https://doi.org/10.1016/s0378-7788(96)00999-1)
- Tewari, M., Chen, F., Kusaka, H., & Miao, S. (2007). *Coupled WRF/Unified Noah/Urban-Canopy Modeling System*. 22.
- Theeuwes, N. E., Steeneveld, G. J., Ronda, R. J., Rotach, M. W., & Holtslag, A. A. M. (2015). Cool city mornings by urban heat. *Environmental Research Letters*, 10(11). <https://doi.org/10.1088/1748-9326/10/11/114022>

- Tian, P., Li, J., Cao, L., Pu, R., Wang, Z., Zhang, H., Chen, H., & Gong, H. (2021). Assessing spatiotemporal characteristics of urban heat islands from the perspective of an urban expansion and green infrastructure. *Sustainable Cities and Society*, 74, 103208. <https://doi.org/10.1016/J.SCS.2021.103208>
- Varnakovid, P., & Ko, H. Y. K. (2023). Urban expansion and urban heat island effects on Bangkok metropolitan area in the context of eastern economic corridor. *Urban Climate*, 52, 101712. <https://doi.org/10.1016/J.UCLIM.2023.101712>
- Wang, Y., Li, Y., Di Sabatino, S., Martilli, A., & Chan, P. W. (2018). Effects of anthropogenic heat due to air-conditioning systems on an extreme high temperature event in Hong Kong. *Environmental Research Letters*, 13(3). <https://doi.org/10.1088/1748-9326/aaa848>
- Webster, D., & McElwee, P. (2009). Urban Adaptation To Climate Change: Bangkok and Ho Chi Minh City As Test Beds. *Fifth Urban Research Symposium*, ii.
- World Bank. (2009). *Climate change impact and adaptation study for Bangkok metropolitan region*.
- World Bank Group. (2021). *Climate Change Knowledge Portal. Thailand Climate Data: Projections*.
- Zheng, Z., Luo, F., Li, N., Gao, H., & Yang, Y. (2024). Impact of Local Climate Zones on the Urban Heat and Dry Islands in Beijing: Spatial Heterogeneity and Relative Contributions. *Journal of Meteorological Research*, 38(1). <https://doi.org/10.1007/s13351-024-3081-6>

Annex: UHII levels in BMA districts

Table A1.1. District mean UHI values in the cold and dry season (December 2019) between 00:00 and 2:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Pom Prap Sattru Phai	4.8	0.72	0.0	0.0	100.0
Samphanthawong	4.8	0.37	0.0	0.0	100.0
Ratchathewi	4.7	1.21	0.0	0.0	100.0
Phra Nakhon	4.7	0.76	0.0	2.0	98.0
Bang Rak	4.7	0.80	0.0	2.6	97.4
Phaya Thai	4.7	1.19	0.0	0.0	100.0
Din Daeng	4.7	2.02	0.0	0.0	100.0
Khlong San	4.6	1.21	0.0	1.9	98.1
Pathum Wan	4.6	0.74	0.0	16.4	83.6
Thon Buri	4.5	1.81	0.0	6.8	93.2
Bangkok Yai	4.5	1.12	0.0	0.0	100.0
Sathon	4.5	1.32	0.0	0.0	100.0
Bang Sue	4.5	2.17	0.0	1.7	98.3
Dusit	4.5	1.43	0.0	15.5	84.5
Bang Phlat	4.4	1.59	0.0	9.3	90.7
Bang Kho Laem	4.3	1.44	0.0	10.1	89.9
Chatuchak	4.3	2.80	0.0	10.7	89.3
Watthana	4.3	1.47	0.0	11.9	88.1
Bangkok Noi	4.2	1.81	0.0	12.4	87.6
Huai Khwang	4.2	1.52	0.0	11.6	88.4
Wang Thonglang	4.2	1.90	0.0	1.9	98.1
Chom Thong	4.2	2.61	0.0	17.9	82.1
Rat Burana	4.0	1.39	0.0	21.6	78.4
Khlong Toei	4.0	1.65	0.0	37.7	62.3
Yan Nawa	4.0	1.35	0.0	23.2	76.8
Lak Si	4.0	1.83	0.0	26.6	73.4
Lat Phrao	4.0	2.08	0.0	46.2	53.8
Phasi Charoen	3.9	2.22	0.0	43.4	56.6
Bang Na	3.8	1.57	0.0	85.4	14.6
Phra Khanon	3.8	1.57	0.0	99.1	0.9
Suan Luang	3.7	2.23	0.4	94.6	4.9
Bang Kapi	3.6	2.57	5.4	71.7	22.9
Bueng Kum	3.5	2.51	0.0	99.5	0.5
Taling Chan	3.3	1.85	1.5	96.0	2.5
Don Mueang	3.2	3.03	0.0	99.1	0.9
Bang Bon	3.1	1.84	7.2	76.5	16.3
Bang Khae	3.0	3.50	13.9	84.1	2.1
Thung Khru	3.0	2.25	5.1	85.4	9.5

Bang Khen	2.9	3.37	16.3	80.8	3.0
Prawet	2.8	3.33	12.1	87.9	0.0
Khan Na Yao	2.8	1.74	15.7	84.3	0.0
Bang Khun Thian	2.8	3.36	2.6	89.9	7.5
Saphan Sung	2.6	1.75	27.1	72.9	0.0
Nong Khaem	2.6	2.83	21.4	78.6	0.0
Sai Mai	2.3	3.80	31.1	68.9	0.0
Thawi Watthana	1.6	1.44	73.2	26.8	0.0
Min Buri	0.9	2.57	80.0	20.0	0.0
Lat Krabang	0.7	3.25	94.1	5.9	0.0
Khlong Sam Wa	0.6	3.81	90.1	9.9	0.0
Nong Chok	-0.2	3.30	100.0	0.0	0.0

Table A1.2. District mean UHI values in the cold and dry season (December 2019) between 18:00 and 20:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Ratchathewi	4.1	1.21	0.0	23.9	76.1
Phaya Thai	4.0	1.19	0.0	31.8	68.2
Din Daeng	4.0	2.02	0.0	23.3	76.7
Pom Prap Sattru Phai	4.0	0.72	0.0	16.7	83.3
Samphanthawong	4.0	0.37	0.0	83.3	16.7
Phra Nakhon	3.9	0.76	0.0	50.0	50.0
Bang Sue	3.9	2.17	0.0	54.2	45.8
Bang Rak	3.9	0.80	0.0	89.5	10.5
Bangkok Yai	3.9	1.12	0.0	94.4	5.6
Pathum Wan	3.9	0.74	0.0	43.8	56.2
Bang Phlat	3.8	1.59	0.0	77.8	22.2
Sathon	3.8	1.32	0.0	100.0	0.0
Dusit	3.8	1.43	0.0	39.8	60.2
Chatuchak	3.8	2.80	0.3	84.9	14.8
Khlong San	3.7	1.21	0.0	100.0	0.0
Wang Thonglang	3.7	1.90	0.0	100.0	0.0
Bangkok Noi	3.7	1.81	2.7	94.7	2.7
Watthana	3.7	1.47	0.0	97.5	2.5
Thon Buri	3.7	1.81	5.4	94.6	0.0
Huai Khwang	3.6	1.52	0.0	98.6	1.4
Lat Phrao	3.6	2.08	0.0	100.0	0.0
Phasi Charoen	3.6	2.22	0.0	100.0	0.0
Lak Si	3.5	1.83	0.0	100.0	0.0
Bang Kho Laem	3.4	1.44	10.1	89.9	0.0
Taling Chan	3.4	1.85	1.2	98.8	0.0
Suan Luang	3.4	2.23	1.3	98.7	0.0
Bang Kapi	3.4	2.57	9.3	90.7	0.0
Bueng Kum	3.4	2.51	8.1	91.9	0.0
Khlong Toei	3.3	1.65	3.3	96.7	0.0
Phra Khanon	3.3	1.57	2.6	97.4	0.0
Chom Thong	3.2	2.61	0.5	99.5	0.0
Bang Na	3.2	1.57	3.4	96.6	0.0
Yan Nawa	3.2	1.35	14.3	85.7	0.0
Bang Khae	3.1	3.50	15.7	84.3	0.0
Don Mueang	2.9	3.03	25.3	74.7	0.0
Rat Burana	2.9	1.39	13.8	86.2	0.0
Bang Khen	2.9	3.37	23.3	76.7	0.0
Khan Na Yao	2.8	1.74	31.9	68.1	0.0
Nong Khaem	2.7	2.83	31.0	69.0	0.0
Saphan Sung	2.6	1.75	33.9	66.1	0.0

Sai Mai	2.6	3.80	30.1	69.9	0.0
Bang Bon	2.6	1.84	32.0	68.0	0.0
Prawet	2.6	3.33	29.8	70.2	0.0
Thawi Watthana	2.0	1.44	56.9	43.1	0.0
Bang Khun Thian	1.6	3.36	71.0	29.0	0.0
Thung Khru	1.5	2.25	61.4	38.6	0.0
Min Buri	1.2	2.57	76.2	23.8	0.0
Khlong Sam Wa	0.9	3.81	86.4	13.6	0.0
Lat Krabang	0.8	3.25	86.6	13.4	0.0
Nong Chok	0.2	3.30	100.0	0.0	0.0

Table A1.3. District mean UHI values in the hot and dry season (March 2020) between 00:00 and 2:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Lak Si	2.0	1.83	12.1	87.9	0.0
Bang Sue	2.0	2.17	32.2	67.8	0.0
Chatuchak	2.0	2.80	45.6	54.4	0.0
Phaya Thai	1.9	1.19	70.6	29.4	0.0
Ratchathewi	1.9	1.21	94.0	6.0	0.0
Din Daeng	1.9	2.02	94.5	5.5	0.0
Pom Prap Sattru Phai	1.8	0.72	100.0	0.0	0.0
Don Mueang	1.8	3.03	40.2	59.8	0.0
Lat Phrao	1.8	2.08	99.5	0.5	0.0
Bang Rak	1.7	0.80	92.1	7.9	0.0
Bang Phlat	1.7	1.59	100.0	0.0	0.0
Wang Thonglang	1.7	1.90	100.0	0.0	0.0
Samphanthawong	1.7	0.37	100.0	0.0	0.0
Dusit	1.7	1.43	90.3	9.7	0.0
Phra Nakhon	1.7	0.76	100.0	0.0	0.0
Bueng Kum	1.7	2.51	100.0	0.0	0.0
Pathum Wan	1.7	0.74	86.3	13.7	0.0
Sathon	1.6	1.32	100.0	0.0	0.0
Bang Khen	1.6	3.37	92.4	7.6	0.0
Bang Kapi	1.6	2.57	100.0	0.0	0.0
Huai Khwang	1.6	1.52	98.6	1.4	0.0
Suan Luang	1.6	2.23	100.0	0.0	0.0
Bangkok Yai	1.6	1.12	100.0	0.0	0.0
Khlong San	1.6	1.21	100.0	0.0	0.0
Watthana	1.5	1.47	99.2	0.8	0.0
Sai Mai	1.5	3.80	83.3	16.7	0.0
Bangkok Noi	1.5	1.81	100.0	0.0	0.0
Phra Khanon	1.5	1.57	100.0	0.0	0.0
Bang Na	1.5	1.57	100.0	0.0	0.0
Phasi Charoen	1.5	2.22	100.0	0.0	0.0
Thon Buri	1.4	1.81	100.0	0.0	0.0
Taling Chan	1.4	1.85	100.0	0.0	0.0
Bang Kho Laem	1.4	1.44	100.0	0.0	0.0
Khan Na Yao	1.4	1.74	100.0	0.0	0.0
Yan Nawa	1.4	1.35	100.0	0.0	0.0
Chom Thong	1.3	2.61	100.0	0.0	0.0
Khlong Toei	1.3	1.65	100.0	0.0	0.0
Bang Khae	1.3	3.50	100.0	0.0	0.0
Rat Burana	1.2	1.39	100.0	0.0	0.0
Saphan Sung	1.2	1.75	100.0	0.0	0.0

Prawet	1.2	3.33	100.0	0.0	0.0
Nong Khaem	1.0	2.83	100.0	0.0	0.0
Bang Bon	1.0	1.84	100.0	0.0	0.0
Thawi Watthana	0.9	1.44	100.0	0.0	0.0
Thung Khru	0.8	2.25	100.0	0.0	0.0
Bang Khun Thian	0.8	3.36	100.0	0.0	0.0
Min Buri	0.6	2.57	100.0	0.0	0.0
Khlong Sam Wa	0.5	3.81	100.0	0.0	0.0
Lat Krabang	0.4	3.25	100.0	0.0	0.0
Nong Chok	-0.2	3.30	100.0	0.0	0.0

Table A1.4. District mean UHI values in the hot and dry season (March 2020) between 18:00 and 20:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Don Mueang	3.7	3.03	0.0	70.5	29.5
Lak Si	3.6	1.83	0.0	99.5	0.5
Sai Mai	3.5	3.80	0.0	91.4	8.6
Bang Khen	3.3	3.37	0.0	100.0	0.0
Bang Sue	3.2	2.17	0.0	100.0	0.0
Chatuchak	3.1	2.80	0.0	100.0	0.0
Bueng Kum	3.1	2.51	0.0	100.0	0.0
Lat Phrao	3.1	2.08	0.0	100.0	0.0
Khan Na Yao	2.8	1.74	0.0	100.0	0.0
Phaya Thai	2.8	1.19	0.0	100.0	0.0
Bang Kapi	2.7	2.57	8.9	91.1	0.0
Wang Thonglang	2.7	1.90	0.0	100.0	0.0
Din Daeng	2.7	2.02	0.0	100.0	0.0
Dusit	2.6	1.43	14.6	85.4	0.0
Ratchathewi	2.6	1.21	0.0	100.0	0.0
Bang Phlat	2.5	1.59	0.0	100.0	0.0
Pom Prap Sattru Phai	2.5	0.72	0.0	100.0	0.0
Pathum Wan	2.4	0.74	20.5	79.5	0.0
Suan Luang	2.4	2.23	2.2	97.8	0.0
Huai Khwang	2.3	1.52	8.2	91.8	0.0
Bang Rak	2.3	0.80	0.0	100.0	0.0
Khlong Sam Wa	2.3	3.81	27.7	72.3	0.0
Phra Nakhon	2.3	0.76	2.0	98.0	0.0
Samphanthawong	2.3	0.37	0.0	100.0	0.0
Saphan Sung	2.2	1.75	27.9	72.1	0.0
Bangkok Noi	2.1	1.81	23.0	77.0	0.0
Sathon	2.0	1.32	40.6	59.4	0.0
Phra Khanon	2.0	1.57	35.9	64.1	0.0
Watthana	2.0	1.47	49.2	50.8	0.0
Min Buri	2.0	2.57	64.7	35.3	0.0
Khlong San	1.9	1.21	57.4	42.6	0.0
Bangkok Yai	1.9	1.12	75.9	24.1	0.0
Taling Chan	1.9	1.85	60.3	39.7	0.0
Bang Na	1.9	1.57	67.4	32.6	0.0
Prawet	1.8	3.33	70.4	29.6	0.0
Thon Buri	1.7	1.81	93.2	6.8	0.0
Bang Kho Laem	1.6	1.44	96.2	3.8	0.0
Thawi Watthana	1.6	1.44	86.1	13.9	0.0
Khlong Toei	1.5	1.65	95.9	4.1	0.0
Phasi Charoen	1.5	2.22	100.0	0.0	0.0

Yan Nawa	1.5	1.35	97.3	2.7	0.0
Nong Chok	1.5	3.30	99.9	0.1	0.0
Bang Khae	1.4	3.50	100.0	0.0	0.0
Lat Krabang	1.4	3.25	92.8	7.2	0.0
Rat Burana	1.2	1.39	100.0	0.0	0.0
Chom Thong	1.1	2.61	100.0	0.0	0.0
Nong Khaem	1.1	2.83	100.0	0.0	0.0
Bang Bon	0.9	1.84	100.0	0.0	0.0
Thung Khru	0.7	2.25	100.0	0.0	0.0
Bang Khun Thian	0.6	3.36	100.0	0.0	0.0

Table A1.5. District mean UHI values in the wet monsoon season (July 2020) between 00:00 and 2:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Pom Prap Sattru Phai	3.3	0.72	0.0	100.0	0.0
Bang Rak	3.2	0.80	0.0	100.0	0.0
Samphanthawong	3.2	0.37	0.0	100.0	0.0
Ratchathewi	3.2	1.21	0.0	100.0	0.0
Phra Nakhon	3.2	0.76	0.0	100.0	0.0
Phaya Thai	3.2	1.19	0.0	100.0	0.0
Pathum Wan	3.1	0.74	0.0	100.0	0.0
Khlong San	3.1	1.21	0.0	100.0	0.0
Din Daeng	3.1	2.02	0.0	100.0	0.0
Sathon	3.1	1.32	0.0	100.0	0.0
Bangkok Yai	3.1	1.12	0.0	100.0	0.0
Thon Buri	3.0	1.81	1.4	98.6	0.0
Bang Sue	3.0	2.17	0.0	100.0	0.0
Dusit	3.0	1.43	0.0	100.0	0.0
Bang Phlat	3.0	1.59	0.0	100.0	0.0
Bang Kho Laem	2.9	1.44	3.8	96.2	0.0
Bangkok Noi	2.9	1.81	6.2	93.8	0.0
Phasi Charoen	2.9	2.22	0.0	100.0	0.0
Chatuchak	2.8	2.80	0.3	99.7	0.0
Chom Thong	2.8	2.61	1.9	98.1	0.0
Wattana	2.8	1.47	0.8	99.2	0.0
Huai Khwang	2.7	1.52	6.8	93.2	0.0
Wang Thonglang	2.7	1.90	0.0	100.0	0.0
Yan Nawa	2.7	1.35	15.2	84.8	0.0
Rat Burana	2.7	1.39	11.2	88.8	0.0
Taling Chan	2.7	1.85	8.3	91.7	0.0
Khlong Toei	2.6	1.65	12.3	87.7	0.0
Bang Khae	2.6	3.50	15.3	84.7	0.0
Lat Phrao	2.6	2.08	6.2	93.8	0.0
Bang Na	2.6	1.57	3.4	96.6	0.0
Lak Si	2.5	1.83	10.6	89.4	0.0
Phra Khanon	2.5	1.57	5.1	94.9	0.0
Bang Bon	2.5	1.84	28.1	71.9	0.0
Suan Luang	2.5	2.23	2.2	97.8	0.0
Nong Khaem	2.4	2.83	29.8	70.2	0.0
Bang Kapi	2.4	2.57	10.9	89.1	0.0
Bang Khun Thian	2.3	3.36	22.9	77.1	0.0
Bueng Kum	2.3	2.51	9.1	90.9	0.0
Thung Khru	2.1	2.25	49.5	50.5	0.0
Thawi Watthana	2.0	1.44	63.3	36.7	0.0

Prawet	2.0	3.33	39.7	60.3	0.0
Don Mueang	1.9	3.03	45.2	54.8	0.0
Bang Khen	1.8	3.37	51.2	48.8	0.0
Khan Na Yao	1.8	1.74	58.7	41.3	0.0
Saphan Sung	1.8	1.75	62.9	37.1	0.0
Sai Mai	1.3	3.80	90.9	9.1	0.0
Lat Krabang	0.7	3.25	100.0	0.0	0.0
Min Buri	0.7	2.57	99.8	0.2	0.0
Khlong Sam Wa	0.4	3.81	100.0	0.0	0.0
Nong Chok	-0.1	3.30	100.0	0.0	0.0

Table A1.6. District mean UHI values in the wet monsoon season (July 2020) between 18:00 and 20:00

DISTRICT	Mean_UHII (Spatial mean)	Population (% of total BMA)	Area(%) UHII<2°C	Area(%) 2°C≤UHII<4°C	Area(%) UHII≥4°C
Din Daeng	2.8	2.02	0.0	100.0	0.0
Ratchathewi	2.7	1.21	0.0	100.0	0.0
Phaya Thai	2.7	1.19	0.0	100.0	0.0
Pom Prap Sattru Phai	2.7	0.72	0.0	100.0	0.0
Bang Rak	2.7	0.80	0.0	100.0	0.0
Bang Sue	2.7	2.17	1.7	98.3	0.0
Wang Thonglang	2.6	1.90	0.0	100.0	0.0
Pathum Wan	2.6	0.74	15.1	84.9	0.0
Samphanthawong	2.6	0.37	0.0	100.0	0.0
Phra Nakhon	2.5	0.76	2.0	98.0	0.0
Chatuchak	2.5	2.80	5.7	94.3	0.0
Bang Phlat	2.5	1.59	5.6	94.4	0.0
Dusit	2.5	1.43	15.5	84.5	0.0
Lat Phrao	2.5	2.08	6.2	93.8	0.0
Huai Khwang	2.5	1.52	11.6	88.4	0.0
Sathon	2.4	1.32	0.0	100.0	0.0
Wattana	2.4	1.47	0.8	99.2	0.0
Bang Kapi	2.4	2.57	10.1	89.9	0.0
Bueng Kum	2.4	2.51	9.1	90.9	0.0
Suan Luang	2.4	2.23	2.2	97.8	0.0
Khlong San	2.3	1.21	1.9	98.1	0.0
Bangkok Yai	2.3	1.12	0.0	100.0	0.0
Bangkok Noi	2.3	1.81	8.8	91.2	0.0
Thon Buri	2.2	1.81	6.8	93.2	0.0
Lak Si	2.2	1.83	12.1	87.9	0.0
Bang Na	2.2	1.57	9.6	90.4	0.0
Phra Khanon	2.2	1.57	15.4	84.6	0.0
Bang Kho Laem	2.1	1.44	15.2	84.8	0.0
Phasi Charoen	2.1	2.22	4.6	95.4	0.0
Yan Nawa	2.0	1.35	22.3	77.7	0.0
Taling Chan	2.0	1.85	20.6	79.4	0.0
Khlong Toei	2.0	1.65	38.5	61.5	0.0
Bang Khen	2.0	3.37	32.2	67.8	0.0
Khan Na Yao	1.9	1.74	37.0	63.0	0.0
Saphan Sung	1.9	1.75	37.5	62.5	0.0
Chom Thong	1.9	2.61	65.6	34.4	0.0
Prawet	1.8	3.33	44.2	55.8	0.0
Bang Khae	1.8	3.50	60.1	39.9	0.0
Rat Burana	1.8	1.39	56.0	44.0	0.0

Don Mueang	1.8	3.03	45.8	54.2	0.0
Sai Mai	1.6	3.80	65.9	34.1	0.0
Nong Khaem	1.6	2.83	95.8	4.2	0.0
Bang Bon	1.6	1.84	83.3	16.7	0.0
Bang Khun Thian	1.4	3.36	97.8	2.2	0.0
Thawi Watthana	1.3	1.44	88.7	11.3	0.0
Thung Khru	1.2	2.25	100.0	0.0	0.0
Min Buri	0.9	2.57	83.7	16.3	0.0
Lat Krabang	0.7	3.25	99.1	0.9	0.0
Khlong Sam Wa	0.5	3.81	92.6	7.4	0.0
Nong Chok	0.0	3.30	100.0	0.0	0.0